DRAFT

KENYA INSTITUTE OF CURRICULUM DEVELOPMENT
A Skilled and Ethical Society

SENIOR SCHOOL CURRICULUM DESIGN

BIOLOGY
GRADE 11

OCTOBER, 2025

First Published in 2025

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ISBN: 978-9914-43-065-3

Published and printed by Kenya Institute of Curriculum Development

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NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. **Foster nationalism and patriotism and promote national unity.**
   Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.

2. **Promote the social, economic, technological and industrial needs for national development.**
   Education should prepare the youth of the country to play an effective and productive role in the life of the nation.
   a) **Social Needs**
   Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution following in the wake of rapid modernization. Education should assist our youth to adapt to this change.
   b) **Economic Needs**
   Education in Kenya should produce citizens with the skills, knowledge, expertise and personal qualities that are required to support a growing economy. Kenya is building up a modern and independent economy which is in need of an adequate and relevant domestic workforce.
   c) **Technological and Industrial Needs**
   Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognizes the rapid industrial and technological changes taking place, especially in the developed world. We can only be part of this development if our education system is deliberately focused on the knowledge, skills and attitudes that will prepare our young people for these changing global trends.

3. **Promote individual development and self-fulfillment**
   Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential interests and abilities. A vital aspect of individual development is the building of character.

4. **Promote sound moral and religious values.**
   Education should provide for the development of knowledge, skills and attitudes that will enhance the acquisition of sound moral values and help children to grow up into self-disciplined, self-reliant and integrated citizens.

5. **Promote social equality and responsibility.**
   Education should promote social equality and foster a sense of social responsibility within an education system which provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability or geographical environment.

6. **Promote respect for and development of Kenya's rich and varied cultures.**
   Education should instill in the youth of Kenya an understanding of past and present cultures and their valid place in contemporary society. Children should be able to blend the best of traditional values with the changing requirements that must follow rapid development in order to build a stable and modern society.

7. **Promote international consciousness and foster positive attitudes towards other nations.**
   Kenya is part of the international community. It is part of the complicated and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities, rights and benefits that this membership entails.

8. **Promote positive attitudes towards good health and environmental protection.**
   Education should inculcate in young people the value of good health in order for them to avoid indulging in activities that will lead to physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM

Senior School is the fourth level of Basic Education in the Competency Based Curriculum (CBC) that the learner shall proceed to after the Pre-Primary School, Primary School and Junior School levels. The essence of Senior School is to offer learners a pre-tertiary/university/ pre-career experience in which the learners have an opportunity to choose pathways, tracks and subjects where they shall have demonstrated interest and/or potential at the earlier levels. Senior School comprises three years of education for learners generally in the 16-18 years age bracket and lays the foundation for further education and training at the tertiary level and the world of work. In the CBC vision, learners exiting this level are expected to be ***engaged, empowered and ethical citizens*** ready to participate in the socio-economic development of the nation.

At this level, the learner shall take seven (7) learning areas as recommended by the *Presidential Working Party on Educational Reforms* (PWPER, 2023). These shall comprise **four compulsory learning areas** namely: English, Kiswahili, Essential Mathematics/Core Mathematics and Community Service Learning (CSL). Learners will choose between Core Mathematics (for learners pursuing the Science Technology Engineering and Mathematics Pathway) and Essential Mathematics (for learners pursuing the non-STEM pathways). The learner will select three learning areas depending on their career choice, aptitude, interest and personality with guidance by the Senior School career teacher.

LEARNING OUTCOMES FOR SENIOR SCHOOL

By the end of Senior School, the learner should be able to:

1. communicate effectively and utilise information and communication technology across varied contexts,
2. apply mathematical, logical and critical thinking skills for problem solving,
3. apply basic research and scientific skills to manipulate the environment and solve problems,
4. exploit individual talents for leisure, self-fulfilment, career growth, further education and training,
5. uphold national, moral and religious values and apply them in day-to-day life,
6. apply and promote health care strategies in day-to-day life,
7. protect, preserve and improve the environment for sustainability,
8. demonstrate active local and global citizenship for harmonious co-existence,
9. demonstrate appreciation of diversity in people and cultures,
10. manage pertinent and contemporary issues responsibly.

TABLE 1: LIST OF SUBJECTS AT SENIOR SCHOOL

Core Subjects	Arts & Sports Science	Social Sciences	Science, Technology, Engineering & Mathematics (STEM)
1. English 2. Kiswahili/KSL 3. Core Mathematics/ Essential Mathematics 4. Community Service Learning (CSL)	5. Sports and Recreation 6. Music and Dance 7. Theatre and Film 8. Fine Arts	9. Literature in English 10. Indigenous Languages 11. Fasihi ya Kiswahili 12. Sign Language 13. Arabic 14. French 15. German 16. Mandarin Chinese 17. Christian Religious Education 18. Islamic Religious Education 19. Hindu Religious Education 20. Business Studies 21. History and Citizenship 22. Geography	23. Biology 24. Chemistry 25. Physics 26. General Science 27. Agriculture 28. Computer Studies 29. Home Science 30. Aviation 31. Building Construction 32. Electricity 33. Metalwork 34. Power Mechanics 35. Woodwork 36. Media Technology 37. Marine and Fisheries Technology

Note:

a) Physical Education (PE) and Information Communication and Technology (ICT) will be offered to all learners to facilitate learning, enjoyment and acquisition of life skills.
b) Every school shall offer Pastoral/Religious Programme of Instruction (P/RPI) to enhance moral, spiritual and character development of the learners.
c) There shall be one (1) lesson for personal or group study to deepen knowledge and enhance skills in areas of choice by the learners.

Table 2 provides guidelines on distribution of lessons at Senior School.

TABLE 2: LESSON DISTRIBUTION AT SENIOR SCHOOL

Subjects	No of Lessons Per Week (40 minutes per lesson)
Core Subjects 4	
1. English	5
2. Kiswahili	5
3. Essential Mathematics/Core Mathematics	5
4. Community Service Learning	3
Elective Subjects	
5. Elective 1	5
6. Elective 2	5
7. Elective 3	5
** Physical Education*	*3*
** ICT Skills*	*2*
**PPI*	*1*
** Learner Personal /Group Study*	*1*
Total Number of Lessons	**40**

ESSENCE STATEMENT

In Senior School, Biology builds on the foundational knowledge and scientific skills acquired in Integrated Science in Junior School. At this level, learners deepen their understanding of the structure and functions of living organisms, their life processes, interaction within the ecosystem, and the continuity of life through reproduction and inheritance. The subject focuses on equipping learners with skills in observation, inquiry, critical thinking, and scientific reasoning. Emphasis is laid on understanding the relevance of Biology in everyday life and its role in addressing real-world issues such as health, food security, and environmental conservation.

Learning in Biology will focus on practical activities, experiments, and scientific investigations. Learners will engage in research, data analysis, and effective communication of findings. The approach will foster the development of core competencies like collaborative learning, critical thinking, and problem-solving. It will also promote learning to learn, where the learners will apply biological concepts to new situations and ensure holistic development. This will not only equip learners for higher learning and the world of work but also for nature-responsible citizenship.

Biology will open up diverse career pathways for learners, especially in fields related to health, agriculture, environmental management, and research. The subject lays a solid foundation for professions such as medicine, nursing, veterinary science, biotechnology, microbiology, environmental science, public health, and forensic science. It also supports innovation and entrepreneurship in areas like pharmaceuticals, food technology, and conservation, thereby contributing to Kenya's socio-economic development and global competitiveness.

SUBJECT GENERAL LEARNING OUTCOMES

By the end of the course, the learner should be able to:

1. develop relevant knowledge, skills and attitudes for further education and for training in biology related scientific fields;
2. demonstrate an understanding of interrelationships among humans, other organisms and their environment and apply the knowledge to conserve nature;
3. describe features of various groups of living organisms and identify unknown organisms using simple biological keys;
4. apply the knowledge gained on human body systems and functions to improve the quality of life for self and the community while enhancing healthy living;

5. design and carry out practical activities and projects that will enable them to understand biological concepts;
6. demonstrate relevant technical skills and scientific knowledge necessary for socio-economic development;
7. demonstrate resourcefulness in designing projects necessary for community service learning;
8. communicate biological information in a precise, clear and logical manner;
9. apply knowledge on plant and animal structure and functions to industrialization, innovation and sustainability of life;
10. use knowledge gained to make informed decisions about scientifically-based personal and societal issues to solve emerging issues in health and environment.

SUMMARY STRANDS AND SUB STRANDS

Strand	Sub Strand
1.0 Cell Biology and Biodiversity	1.1 Taxonomy I
	1.2 Ecology
	1.3 Taxonomy II
	1.4 Cell Division
2.0 Anatomy and Physiology of Plants	2.1 Reproduction in Plants
	2.2 Growth and Development in Plants
	2.3 Excretion in Plants
3.0 Anatomy and Physiology of Animals	3.1 Reproduction in Animals
	3.2 Growth and Development in Animals
	3.3 Excretion and Homeostasis in Animals

STRAND 1.0: CELL BIOLOGY AND BIODIVERSITY

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Cell Biology and Biodiversity	**1.1 Taxonomy I** • Meaning and significance of taxonomy • Taxonomic units and nomenclature • Five Kingdoms of classification- Monera, Protoctista, Fungi, Plantae, and Animalia *(General characteristics of the five kingdoms; Economic importance of Kingdoms Monera, Protoctista and Fungi)*	By the end of the sub strand, the learner should be able to: a) explain the significance of classification of living organisms in nature, b) illustrate the hierarchy of the major taxonomic units of classification, c) describe the principles of binomial nomenclature in classification of living organisms, d) classify organisms into the five taxonomic Kingdoms, e) appreciate the importance of taxonomy in biodiversity conservation.	The learner is guided to: • search for information on the meaning and significance of taxonomy, and share with peers, • design charts to show the hierarchy of major taxonomic units of classification, and present to peers, • discuss the principles of binomial nomenclature with peers, • name common organisms in the local environment using principles of binomial nomenclature, • explore the local environment/print media/non print media to observe and classify organisms into Kingdom Monera, Protoctista, Fungi, Plantae, and Animalia (*Note: Observe road safety during the activity*), • use hand lenses and light microscopes to observe specimens in Kingdoms Monera, Protoctista, and Fungi, • explore the community to appreciate the economic importance of organisms in Kingdoms Monera, Protoctista, and Fungi, and discuss with peers.	How are living organisms classified?

Core Competencies to be developed:
- Communication and Collaboration: The learner contributes to group decision-making while exploring the local environment/print media/non print media to observe and classify organisms.
- Self-efficacy: The learner's confidence is enhanced as they design charts to show the hierarchy of major taxonomic units of classification and present them to peers.

Values
- Respect: The learner appreciates diverse opinions while discussing the principles of binomial nomenclature with peers.
- Responsibility: The learner observes safety precautions while naming common organisms in the local environment using principles of binomial nomenclature.

Pertinent and Contemporary Issues (PCIs)
- Safety and Security: The learner observes road safety as they explore the local environment to observe and classify organisms.
- Environmental Conservation: The learner appreciates the need to conserve biodiversity while discussing the economic importance of organisms in Kingdoms Monera, Protoctista, and Fungi.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Cell Biology and Biodiversity	**1.2 Ecology** • Ecosystems *(forest, grassland, desert, and aquatic)* • Adaptations of organisms to different habitats *(xerophytes, mesophytes, hydrophytes, halophytes, desert and aquatic animals)* • Characteristics of populations and methods of estimating population size *(direct count and sampling methods - line transect, belt transect, quadrat, capture-recapture)* • Factors that affect populations in ecosystems *(biotic and abiotic factors)*, • Nutrient cycles and their importance in the ecosystems *(nitrogen and carbon cycles)* • Effects of environmental pollution *(include global warming and climate change)*	By the end of the sub strand, the learner should be able to: a) describe the characteristics of different ecosystems in nature, b) explain the adaptations of organisms to different habitats, c) describe factors affecting population sizes in ecosystems, d) estimate population sizes of organisms in ecosystems, e) illustrate the nitrogen and carbon cycles in ecosystems, f) describe the effects of pollution in the environment, g) appreciate the link between environmental pollution and climate change.	The learner is guided to: • explore available resources for information on terminologies used in ecology *(habitat, niche, population, community, biome, ecosystem, biomass, carrying capacity, ecological succession)*, and share with peers, • explore the environment and other resources to observe the characteristics of different ecosystems *(forest, grassland, desert, and aquatic; - include climatic conditions, physiographic, and biotic factors of the ecosystems)*. Model the ecosystems, and discuss with peers, • observe specimens and multimedia (photographs, videos, animations) showing various adaptations of organisms to different habitats, and discuss with peers, • role-play and simulate characteristics of a population *(density, dispersion, and growth)* and factors that affect populations in ecosystems, • use direct count and sampling methods *(line transect, belt transect, quadrat, capture-recapture)* to estimate the number of organisms in given ecosystems, • draw charts and/or make models to illustrate nitrogen and carbon cycles, and present to peers, • explore the community and other resources to observe and discuss the effects of environmental pollution and suggest control measures, • brainstorm on the causes and impacts of climate change, record the main points, and present to peers, • make posters to sensitise the community about the effects of environmental pollution and their mitigation measures.	How do organisms interact with their environment?

Core Competencies to be developed:
- Citizenship: The learner solves problems in the local community as they make posters to sensitise the community about the effects of environmental pollution.
- Self-efficacy: The learner communicates clearly with others while brainstorming on the causes and impacts of climate change.

Values
- Integrity: The learner utilises resources prudently while making charts and models to illustrate nitrogen and carbon cycles.
- Patriotism: The learner demonstrates love for the country while sensitising the community about the effects of environmental pollution.

Pertinent and Contemporary Issues (PCIs)
- Waste Management: The learner appreciates the importance of proper waste management as they sensitise the community about the effects of environmental pollution and their mitigation measures.
- Prevention of Global Warming: The learner appreciates the need to reduce greenhouse gas emissions while observing and discussing the effects of environmental pollution.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Cell Biology and Biodiversity	**1.3 Taxonomy II** • Kingdom Plantae *(general characteristics of organisms in Divisions Bryophyta, Pteridophyta, and Spermatophyta; include sub-divisions and Classes of Spermatophyta)* • Kingdom Animalia *(general characteristics of organisms in Chordata and Arthropoda up to class level; include economic importance of arthropods)* • Dichotomous keys *(principles of construction, identification of organisms, and practical applications)* • Viruses *(general characteristics, conditions for growth, and economic importance)*	By the end of the sub strand, the learner should be able to: a) outline the characteristics of organisms in the divisions of Kingdom Plantae, b) classify organisms into phyla and classes of Kingdom Animalia, c) construct dichotomous keys to identify plants and animals, d) relate viruses to their economic importance in health, agriculture, and biotechnology, e) appreciate the diversity of plants and animals in nature.	The learner is guided to: • collaboratively search for information on general characteristics of organisms in phyla/divisions Kingdom Animalia and Plantae, • explore the local environment to observe characteristics of members of the various divisions in Kingdom Plantae, and discuss with peers, • use distinguishing features to group specimens into sub-divisions of Spermatophyta (Gymnospermae and Angiospermae), and discuss with peers, • collect and observe specimens of monocot and dicot plants and discuss their distinguishing characteristics, • observe and group animals into Phylum Chordata and Arthropoda using distinguishing characteristics, discuss with peers, • use distinguishing features to place organisms in Phylum Chordata and Arthropoda into their taxonomic classes, • construct and use dichotomous keys to identify various plants and animals, • brainstorm with peers on the characteristics of viruses and their economic importance. **Projects** Use locally available materials and organisms within the school/home environment to: • Prepare herbarium specimens of various plant species for future reference during biological studies. • Mount animal specimens/their images on a soft board/wooden surface/skin/carton or any available surface. *(Safe and appropriate preservatives may be used in the preservation of some specimens as may be necessary)*.	1. How are plants and animals classified? 2. How are viruses important in nature?

Core Competencies to be developed:
- Creativity and Imagination: The learner experiments with varied ideas while preparing a herbarium of plant specimens.
- Communication and Collaboration: The learner speaks clearly to peers while discussing distinguishing characteristics of various specimens.

Values
- Respect: The learner is considerate of others' opinions while brainstorming on the characteristics of viruses and their economic importance.
- Responsibility: The learner cares for the environment while collecting specimens of monocot and dicot plants.

Pertinent and Contemporary Issues (PCIs)
- Environmental Conservation: The learner conserves biodiversity by responsibly collecting organisms for the preparation of herbarium specimens.
- Animal Welfare: The learner handles animals with care while observing and grouping them into Phylum Chordata and Arthropoda using distinguishing characteristics.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Cell Biology and Biodiversity	**1.4 Cell Division** • Meaning of cell division and structure of chromosomes • Types of cell division *(mitosis and meiosis)* • Significance of cell division in organisms *(include tissue culture)* • Budding and binary fission	By the end of the sub strand, the learner should be able to: a) explain the significance of mitosis and meiosis in organisms, b) illustrate the stages of mitosis and meiosis in cells, c) distinguish between budding and binary fission in relation to cell division, d) recognise the importance of cell division in reproduction, growth, and development.	The learner is guided to: carry out practical activities to discuss the meaning • use models and drawings to explain the meaning and significance of cell division and illustrate the structure of chromosomes. Discuss with peers, • watch animations showing various stages of mitosis and meiosis, and discuss with peers, • use models and charts to illustrate the behaviour of chromosomes at various stages of cell division *(mitosis and meiosis)*, and present to peers, • use a light microscope to observe stages of meiosis and mitosis in living tissues, • model budding in yeast and binary fission in bacteria and discuss with peers.	How is cell division important?

Core Competencies to be developed:
- Creativity and Imagination: The learner innovatively models budding in yeast and binary fission in bacteria.
- Digital Literacy: The learner enhances computing skills while watching animations showing various stages of mitosis and meiosis.

Values
- Unity: The learner collaborates with others while using a light microscope to observe stages of meiosis and mitosis in living tissues.
- Respect: The learner appreciates others' efforts while presenting illustrations to show the behaviour of chromosomes in various stages of cell division.

Pertinent and Contemporary Issues (PCIs)
- Safety and Security: The learner observes safety precautions while using a light microscope to observe stages of meiosis and mitosis in living tissues.
- Communicable Diseases: The learner recognizes the importance of healthy practices for prevention of communicable diseases while modelling budding in yeast and binary fission in bacteria.

Suggested Assessment Rubric

Indicator \ Level	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to illustrate the hierarchy of the major taxonomic units of classification	Illustrates the hierarchy of all the major taxonomic units of classification with relevant examples of organisms for each taxon	Illustrates the hierarchy of all the major taxonomic units of classification	Illustrates the hierarchy of 4-6 major taxonomic units of classification	Illustrates the hierarchy of 1-3 major taxonomic units of classification
Ability to estimate population sizes of organisms in ecosystems	Estimates population sizes of organisms in ecosystems using all 5 methods with clear justification for each method	Estimates population sizes of organisms in ecosystems using all 5 methods	Estimates population sizes of organisms in ecosystems using 2-4 methods	Estimates population sizes of organisms in ecosystems using 1 method
Ability to classify organisms into phyla and classes of Kingdom Animalia	Accurately classifies organisms into all phyla and classes of Kingdom Animalia citing distinguishing features	Accurately classifies organisms into all phyla and classes of Kingdom Animalia	Accurately classifies organisms into either phyla or classes of Kingdom Animalia	Classifies some organisms into either phyla or classes of Kingdom Animalia
Ability to illustrate the stages of mitosis and meiosis in cells	Illustrates the stages of mitosis and meiosis in cells, explaining the significance of each stage	Illustrates the stages of mitosis and meiosis in cells	Illustrates the stages of either mitosis or meiosis in cells	Illustrates some stages of either mitosis or meiosis in cells

STRAND 2.0: ANATOMY AND PHYSIOLOGY OF PLANTS

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Anatomy and Physiology of Plants	**2.1 Reproduction in Plants** • Asexual reproduction *(Sporulation and vegetative propagation - include specialised vegetative structures such as runners, bulbs, corms, rhizomes, offshoots, tubers and cuttings)* • Sexual reproduction in flowering plants *-Identification of flowers based on the position of the ovary, symmetry, inflorescence, presence of male and female parts -Mechanisms and features that hinder self- pollination and self-fertilisation -Double fertilisation and its significance -Seed and fruit development (include placentation and classification of fruits)*	By the end of the sub strand, the learner should be able to: a) describe asexual reproduction in plants, b) identify flowers based on their floral parts, c) illustrate mechanisms and features that hinder self-pollination and self-fertilisation in plants, d) describe the process of double fertilisation in flowering plants, e) acknowledge the importance of sexual and asexual reproduction in plants.	The learner is guided to: • discuss asexual reproduction in plants with peers, • observe vegetative propagation of plants in the locality or digital resources, and discuss with peers, • observe specimens of flowers from different plants and classify the flowers based on floral parts, • use models and charts to illustrate mechanisms and features that hinder self-pollination and self-fertilisation in plants, and present to peers, • watch animations illustrating double fertilisation and formation of seeds and fruits. Discuss with peers, • examine, draw and label sections of fruits, and discuss their placentation and classification, • brainstorm on the advantages and disadvantages of sexual and asexual reproduction in plants.	1. What is the significance of reproduction in plants? 2. How does reproduction occur in plants?

Core Competencies to be developed:
- Communication and Collaboration: The learner cooperates with others while observing and classifying specimens of flowers from different plants.
- Critical Thinking and Problem Solving: The learner analyses causes and effects while brainstorming on the advantages and disadvantages of sexual and asexual reproduction in plants.

Values
- Social Justice: The learner shares resources equitably while examining, drawing, and labelling sections of fruits.

- Love: The learner prioritises others' interests while using models and charts to illustrate mechanisms and features that hinder self- pollination and self-fertilisation in plants.

Pertinent and Contemporary Issues (PCIs)
- Environment and Technology: The learner appreciates the use of technology in managing the environment while watching animations illustrating double fertilisation and formation of seeds and fruits.
- Online Safety: The learner explores safe internet sites while watching animations illustrating double fertilisation and formation of seeds and fruits.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Anatomy and Physiology of Plants	**2.2 Growth and Development in Plants** • Meaning of growth and development • Structure of the seed *(monocots and dicots)* • Seed dormancy *(meaning, significance, causes, and ways of breaking seed dormancy)* • Germination *(process, types, and conditions necessary for germination)* • Primary and secondary growth • Measurement of growth and growth curves • Role of hormones in plant growth and development *(include apical dominance)*	By the end of the sub strand, the learner should be able to: a) distinguish between growth and development in living organisms, b) relate structures of monocot and dicot seeds to their functions, c) outline causes and ways of breaking dormancy in seeds, d) investigate conditions necessary for seed germination, e) investigate types of germination in plants, f) illustrate primary and secondary growth in flowering plants, g) demonstrate the role of hormones in plant growth and development, h) appreciate the importance of growth and development in plants.	The learner is guided to: • brainstorm on the meaning and significance of growth and development in living organisms, • observe, draw, and label the structures of monocot and dicot seeds, and discuss the functions of the structures with peers, • examine seed specimens or photographs to identify causes of seed dormancy, and discuss with peers, • carry out activities to demonstrate ways of breaking seed dormancy, • use locally available seeds and materials to investigate conditions necessary for seed germination, and discuss with peers, • carry out experiments to investigate different types of germination (hypogeal and epigeal) using locally available seeds, and present their work to peers, • observe sections of stems under a light microscope or photomicrographs showing primary and secondary growth in plants. Draw and label the observed sections and present to peers, • use models and drawings to illustrate stages of secondary growth in a woody dicotyledonous plant and present to peers, • plant locally available seeds and use the seedlings to determine regions of growth, and discuss with peers, • record experimental data (such as height of seedling or width of stem) from a growing seedling, draw growth curves, and discuss with peers, • watch animations illustrating the effect of plant growth hormones and their application in agriculture. **Project:** Establish a tree nursery by generating roots on stem cuttings using natural sources of hormones (such as *bulb onions, Aloe spp., cinnamon)*. Sell the seedlings to members of the community at a subsidized price.	How does growth occur in plants?

Core Competencies to be developed:
- Citizenship: The learner solves problems in the community by sharing the seedlings from the established tree nursery.
- Learning to Learn: The learner reflects on their own work and adjusts accordingly while carrying out activities to demonstrate ways of breaking seed dormancy.

Values
- Unity: The learner works harmoniously with others while carrying out experiments to investigate conditions necessary for seed germination.
- Patriotism: The learner demonstrates devotion to the country/community while sharing seedlings with the school community for environmental conservation.

Pertinent and Contemporary Issues (PCIs)
- Financial Literacy: The learner explores ways of generating income while selling seedlings to members of the community.
- Prevention of Global Warming: The learner mitigates the effects of global warming by establishing a tree nursery and distributing seedlings in the community.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Anatomy and Physiology of Plants	**2.3 Excretion in Plants** • Meaning and importance of excretion *(include reasons why plants lack complex excretory structures)* • Methods of excretion in plants • Economic importance of plant excretory products	By the end of the sub strand, the learner should be able to: a) describe ways through which plants eliminate waste products, b) demonstrate ways through which plants eliminate waste products, c) outline the economic importance of plant excretory products, d) recognise the importance of excretory products in plants.	The learner is guided to: • discuss the meaning and importance of excretion, and brainstorm on reasons why plants lack complex excretory structures, • take a nature walk in the school locality to observe ways through which plants eliminate wastes and discuss *(Transpiration, diffusion, deposition, organ/tissue fall, exudation and guttation)*, • carry out experiments to demonstrate ways through which plants eliminate waste products, and report findings to peers, • search for information on plant excretory products and their economic importance, and discuss with peers. **Project:** • Conduct interviews with members of the local communities about the uses of various plant products. Document the plant products and their uses and make a portfolio showing photographs of the plants and their excretory products.	1. How do plants eliminate waste products? 2. How do humans use plant excretory products?

Core Competencies to be developed:
- Learning to Learn: The learner shares information with others discussing plant excretory products and their economic importance.
- Self-efficacy: The learner enhances self-confidence while carrying out experiments to demonstrate ways through which plants eliminate waste products.

Values
- Respect: The learner is considerate of others' opinions while brainstorming on reasons why plants lack complex excretory structures.
- Integrity: The learner demonstrates honesty while reporting findings on ways through which plants eliminate waste products.

Pertinent and Contemporary Issues (PCIs)
- Drug and Substance Abuse: The learner is sensitised about the effects of substance abuse while conducting interviews with members of the local communities about the uses of various plant products.
- Financial Literacy: The learner appreciates the commercial value of plant excretory products while exploring and brainstorming on their economic importance.

Suggested Assessment Rubric

Indicator \ Level	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to illustrate mechanisms and features that hinder self- pollination and self-fertilisation in plants	Illustrates all mechanisms and features that hinder self-pollination and self-fertilisation in plants with relevant examples of plants	Illustrates all mechanisms and features that hinder self-pollination and self-fertilisation in plants	Illustrates 3-5 mechanisms and features that hinder self-pollination and self-fertilisation in plants	Illustrates 1-2 mechanisms and features that hinder self-pollination and self-fertilisation in plants
Ability to investigate conditions necessary for seed germination	Investigates 3 conditions necessary for seed germination and accounts for the observations	Investigates 3 conditions necessary for seed germination	Investigates 2 conditions necessary for seed germination	Investigates 1 condition necessary for seed germination
Ability to illustrate primary and secondary growth in flowering plants	Illustrates primary and secondary growth in flowering plants with illustrations	Illustrates primary and secondary growth in flowering plants	Illustrates either primary or secondary growth in flowering plants	Partially illustrates either primary or secondary growth in flowering plants
Ability to describe ways through which plants eliminate waste products	Describes all ways through which plants eliminate waste products, citing examples of plants	Describes all ways through which plants eliminate waste products	Describes 2-3 ways through which plants eliminate waste products	Describes 1 way through which plants eliminate waste products

STRAND 3.0: ANATOMY AND PHYSIOLOGY OF ANIMALS

Strand	Sub Strand	Specific Learning outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Anatomy and Physiology of Animals	**3.1 Reproduction in Animals** • External fertilisation in animals • Types of internal fertilisation *(viviparous, oviparous and ovoviviparous animals)* • Structure of the male and female reproductive systems • Menstrual cycle • Fertilisation, Pregnancy, birth and lactation *(include the structure of ovum and sperm cell, and role of hormones)* • Role of biotechnology in human reproduction *(surrogacy, sperm bank, in-vitro fertilisation)*	By the end of the sub strand, the learner should be able to: a) describe external fertilisation in animals, b) describe types of internal fertilisation in animals, c) relate the structure of the male and female reproductive systems to their functions, d) describe the process of fertilisation, pregnancy, and birth in humans, e) describe the role of hormones in human reproduction, f) appreciate the role of biotechnology in addressing human reproduction health.	The learner is guided to: • observe external fertilisation in animals *(amphibians and fish)* in their habitats *(ponds and aquaria)* or from other available resources, and discuss with peers, • search for information from available sources on types of internal fertilisation and share with peers, • make classification charts to group animals based on types of internal fertilisation, and present to peers, • use models and charts to illustrate the structure of the male and female reproductive systems and discuss their adaptations with peers, • dissect a small mammal to observe the reproductive system. Draw and label, • use drawings to illustrate the human sperm cell and ovum, and present to peers, • use digital devices to watch animations showing fertilisation, pregnancy, placenta and birth in humans; discuss with peers, • learn about the stages of pregnancy and the process of childbirth from a resource person (midwife or nurse) to help mitigate postnatal maternal health issues, • use charts to illustrate the role of hormones in the human sexual cycle, pregnancy, birth and lactation, discuss with peers, • brainstorm on the effects of undersecretion and oversecretion (imbalance) of reproductive hormones, • debate on the role of biotechnology in human reproduction, note down the main points, and present to peers, • Debate on the importance of joint reproductive health choices made by both male and female parents.	How do animals reproduce?

Core Competencies to be developed:
- Critical thinking and Problem Solving: The learner gives objective ideas to solve reproductive health problems while debating on the role of biotechnology in human reproduction.
- Learning to Learn: The learner seeks information from the resource person while learning about the stages of pregnancy and the process of childbirth from a resource person.

Values:
- Social Justice: The learner shares responsibilities fairly while making classification charts to group animals based on types of internal fertilisation.
- Peace: The learner resolves differences amicably while debating on the role of biotechnology in human reproduction.

Pertinent and Contemporary Issues (PCIs):
- Menstrual Health Disorders Management: The learner is sensitised on human menstrual health while illustrating the role of hormones in the human sexual cycle.
- Human Sexuality: The learner understands common myths and misconceptions around human sexuality while illustrating the role of hormones in the human sexual cycle.

Strand	Sub strand	Specific Learning outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Anatomy and Physiology of Animals	**3.2 Growth and Development in Animals** • Metamorphosis *(complete and incomplete metamorphosis in insects)* • Role of growth hormones in insects • Growth curves in animals' *(continuous and intermittent growth)*	By the end of the sub strand, the learner should be able to: a) distinguish between complete and incomplete metamorphosis in insects, b) describe the role of hormones in growth and development in insects, c) analyse growth curves in animals, d) appreciate the importance of growth and development in animals.	The learner is guide to: • search for information from available resources on types of growth in animals, and discuss with peers, • use available resources to model life cycles of named insects and make presentations to peers, • plot graphs to illustrate complete and incomplete metamorphosis in insects, and present to peers, • search for information on insect hormones and their roles in growth and development. Share with peers, • watch animations illustrating the effect of hormones on insect growth and development, • collect data *(height, length, or mass)* of animals, plot and interpret growth curves *(intermittent and sigmoid)*. Calculate growth rates at different growth phases.	How does growth in animals occur?

Core Competencies to be developed
- Digital Literacy: The learner appropriately selects animations illustrating the effect of hormones on insect growth and development.
- Creativity and Imagination: The learner adjusts and rethinks their ideas while modelling life cycles of different insects.

Values
- Love: The learner handles insects with compassion as they collect data on animals to calculate growth rates at different growth phases.
- Unity: The learner collaborates with others while plotting graphs to illustrate complete and incomplete metamorphosis in insects.

Pertinent and Contemporary Issues (PCIs)
- Good governance: The learner demonstrates principles of leadership while working with peers to search for information on insect hormones and their roles in growth and development.
- Animal Welfare: The learner treats animals with care while collecting data for plotting growth curves in animals.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Anatomy and Physiology of Animals	**3.3 Excretion and Homeostasis in Animals** • Excretory organs and products in animals *(insects, birds, freshwater and marine fish)* • Internal structure of the mammalian kidney *(cortex, medulla, pelvis, structure and function of the nephron)* • Process of urine formation in mammals, • Internal structure of the mammalian skin, • Role of the liver and skin in excretion, • Homeostatic regulation in humans *(osmoregulation, thermoregulation, blood sugar regulation – include diabetes mellitus and diabetes insipidus)* • Liver diseases and disorders. *(cirrhosis, hepatitis, fatty liver disease).*	By the end of the sub strand, the learner should be able to: a) relate excretory structures to their products in insects, birds, and fish, b) relate the internal structure of the mammalian kidney to its function, c) describe the process of urine formation in mammals, d) relate the structure of the mammalian liver and skin to their excretory functions, e) illustrate homeostatic regulation in humans, f) reflect on the importance of excretion and homeostasis in animals.	The learner is guided to: • explore available sources for information on excretory structures and products in insects, birds and fish, • dissect a mammalian kidney to observe and draw the internal structure. Discuss with peers. (The kidney may be obtained from a locally available animal), • model and label the internal structure of a mammalian kidney and relate it to its functions. Present to peers, • use digital devices to watch animations illustrating the process of urine formation in mammals and discuss with peers, • observe photomicrographs or illustrations of the internal structure of the mammalian liver and skin and relate them to their functions, • use charts to illustrate mechanisms of homeostasis in humans (osmoregulation, blood sugar regulation, thermoregulation-cover homeostatic feedback mechanisms), • listen to a resource person to learn about liver diseases and disorders in humans, and discuss with peers, • observe and/or discuss how animals in the locality respond to changes in the environment (*include basking, hibernation, burrowing, panting, aestivation*).	1. How do animals get rid of waste products? 2. How do animals cope with environmental changes?

Core Competencies to be developed:
- Critical Thinking and Problem Solving: The learner asks questions to be clear on what they need while discussing how animals in the locality respond to changes in the environment.
- Digital Literacy: The learner uses digital technology effectively while using digital devices to watch animations on the process of urine formation in mammals.

Values
- Social Justice: The learner shares resources equitably while modelling and labelling the internal structure of a mammalian kidney.
- Responsibility: The learner engages in assigned roles and duties while dissecting a mammalian kidney to observe the internal structure.

Pertinent and Contemporary Issues (PCIs)
- Environmental Conservation: The learner observes laboratory safety while dissecting the mammalian kidney to observe internal structure.
- Waste Management: The learner appreciated the need to conserve biodiversity while observing and discussing how animals in the locality respond to changes in the environment.

Suggested Assessment Rubric

Indicator \ Level	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to relate the structure of the male and female reproductive systems to their functions	Relates the structure of the male and female reproductive systems to their functions with appropriate illustrations	Relates the structure of the male and female reproductive systems to their functions	Relates the structure of either the male or female reproductive systems to their functions	Relates the structure of either the male or female reproductive systems to some of their functions
Ability to describe the role of hormones in growth and development in insects	Describes the role of all hormones in growth and development in insects, citing the mode of action	Describes the role of all hormones in growth and development in insects	Describes the role of 2 hormones in growth and development in insects	Describes the role of 1 hormone in growth and development in insects
Ability to relate the structure of the mammalian liver and skin to their excretory functions	Relates the structure of the mammalian liver and skin to their excretory function, with relevant illustrations	Relates the structure of the mammalian liver and skin to their excretory functions	Relates the structure of either the mammalian liver or skin to their excretory functions	Relates the structure of either the mammalian liver or skin to some of their excretory functions
Ability to relate the internal structure of the mammalian kidney to its function	Relates all internal structures of the mammalian kidney to its function with appropriate illustrations	Relates all internal structure of the mammalian kidney to its function	Relates 3-4 internal structures of the mammalian kidney to their functions	Relates 1-2 internal structure of the mammalian kidney to their functions

APPENDIX: SUGGESTEDD METHODS, LEARNING RESOURCES AND NON-FORMAL ACTIVITIES

Assessment Methods in Science	Learning Resources	Non-Formal Activities
• Reflections • Game Playing • Pre-Post Testing • Model Making • Explorations • Experiments • Investigations • Conventions, Conferences and Debates • Teacher Observations • Project • Journals • Portfolio • Oral or Aural Questions • Learner's Profile • Written Tests • Anecdotal Records	• Laboratory Apparatus and Equipment • Textbooks • Models • Digital media (Radio and TV education programmes, Kenya education cloud and OERs) • Print media (charts, pictures, journals, magazines) • Digital Devices • Software • Recordings • Resource persons	• Visit the science historical sites. • Use digital devices to conduct scientific research. • Organising walks to have live learning experiences. • Developing simple guidelines on how to identify and solve some community problems. • Conducting science document analysis. • Participating in talks by resource persons on science concepts. • Participating in science clubs and societies. • Attending and Participating in Science and Engineering fairs. • Organising and participating in exchange programs. • Making oral presentations and demonstrations on science issues.